VoiceForAll

Bidirectional AI-Powered Sign Language Translation System

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Abstract

VoiceForAll is a real-time American Sign Language recognition system designed to bridge the communication barrier between deaf signers and hearing individuals. The system uses MediaPipe Holistic to extract hand landmarks from a standard laptop camera, a Bidirectional LSTM model trained on the PopSign ASL v1.0 dataset [1] to classify signs, and a locally-running large language model with confidence-based filtering to convert recognised signs into grammatically correct English sentences, which are then spoken aloud via text-to-speech, running entirely offline with no specialist hardware or internet connection required.

The system was developed iteratively across six versions, systematically testing different architectures and feature representations. The central finding of the project is that reducing input features from 225 landmarks to 63 hand-only landmarks produced a greater improvement in real-world performance than any architectural change, eliminating a persistent gap between laboratory test accuracy and live webcam performance that reached as high as 53 percentage points in earlier versions.

The final system recognises 248 ASL signs with 72.82% test accuracy. Real-world accuracy reaches 87.5% on visually distinct signs, with an overall live accuracy of 63.7% across the full 248-sign vocabulary, a gap explained by a small subset of signs that are inherently indistinguishable without facial landmark data, and transparently documented in the evaluation. This work demonstrates that accessible, privacy-preserving sign language translation is achievable on standard consumer hardware, contributing towards more inclusive everyday communication for the deaf community.

1 Introduction and Motivation

The motivation for this project is personal. Growing up, I watched my cousin, who is deaf, sit alone at family gatherings because no one around him knew sign language, and sign language was the only way he could communicate. During a recent visit, I wanted to have a conversation with him, but found myself reduced to guessing at what he was trying to tell me. Despite his best efforts, the barrier between us felt insurmountable. That experience made one thing clear: the problem is not with the person who is deaf, it is with the tools available to bridge the communication gap.

This experience is far from unique. According to the World Federation of the Deaf, over 70 million people worldwide use sign language as their primary means of communication [2], yet the vast majority of hearing people have no knowledge of it whatsoever. In the United Kingdom alone, an estimated 151,000 people use British Sign Language, of whom 87,000 are Deaf [3, 4]. The communication barrier this creates is not a minor inconvenience; it limits access to healthcare, education, employment, and everyday social interaction. For many deaf individuals, this barrier is present in every interaction outside of their immediate community.

Despite the scale of this problem, existing solutions remain inadequate for everyday communication. A common assumption is that deaf individuals can simply resort to written language, typing or texting, as an alternative. In reality, for a significant proportion of the deaf community, written English is a second language. Sign languages such as ASL and BSL have their own grammar, syntax, and sentence structure entirely distinct from English, and many deaf individuals, particularly those who have been signing from birth, find written communication effortful and slow [4]. Professional sign language interpreters exist and remain the gold standard for high-stakes environments such as hospitals and courtrooms, but they are critically scarce. According to the National Registers of Communication Professionals working with Deaf and Deafblind People (NRCPD), there are only 908 registered sign language interpreters and a further 234 trainees serving a community of over 87,000 deaf BSL signers [3], meaning interpreters are expensive, require advance booking, and are simply unavailable for the spontaneous, everyday interactions that hearing people take for granted, such as asking a pharmacist a question, speaking to a colleague, or ordering food. Existing digital tools partially address the problem but none solve it completely: speech-to-text applications help deaf people understand hearing speakers but do nothing to help them be understood in return, while sign language translation tools either require specialist hardware, depend on an internet connection, or lack the accuracy needed for reliable real-world use.

VoiceForAll is a real-time sign language translation system designed to address this gap. Using only a standard laptop camera, the system detects and tracks hand movements using MediaPipe [5], recognises American Sign Language signs using a Bidirectional LSTM model [6] trained on the PopSign ASL v1.0 dataset [1], converts the recognised signs into grammatically correct

English sentences using a locally-running large language model, and speaks them aloud via text-to-speech, all without any internet connection or specialist hardware. The primary contribution of this project is the sign-to-speech pipeline, which has been fully implemented, trained, and evaluated across 248 ASL signs. Text-to-sign, the reverse direction allowing a hearing person to type a word and see it signed by a real human on screen, is identified as the natural next step and a direction for future development, informed by feedback from Professor Adam Schembri, a world-leading scholar in Sign Language Studies at the University of Birmingham. This report documents the motivation behind the system, the iterative development process across six versions, the architectural decisions that led to the final design, and a critical evaluation of the real-world performance of the system.

2 Background and Related Work

2.1 Sign Language as a Linguistic System

Sign languages are complete, natural human languages with their own grammar, syntax, vocabulary, and linguistic structure entirely independent of the spoken languages used in the same country [7]. American Sign Language (ASL) and British Sign Language (BSL), for example, are mutually unintelligible despite both being used in English-speaking countries, differing fundamentally in handshape, movement, and grammatical structure [8]. Unlike spoken languages, sign languages are three-dimensional. Meaning is conveyed not only through hand shapes but also through the location of the hands relative to the body, the movement and direction of the hands, facial expressions, and mouth patterns [7]. Facial grammar in particular plays a critical role: raised eyebrows indicate a yes/no question, furrowed eyebrows indicate a wh-question, and mouth movements can modify the meaning of a sign entirely [9]. This linguistic richness means that sign language recognition is a fundamentally harder problem than it might appear. A system that tracks only hand position is capturing an incomplete picture of the language, and a system trained on one sign language cannot be directly applied to another [1].

2.2 Early Approaches to Sign Language Recognition

Research into automated sign language recognition dates back to the early 1990s [10]. The earliest systems relied on instrumented gloves, wearable devices embedded with sensors that directly measured finger bend angles, wrist orientation, and hand position in three-dimensional space [10]. While these systems achieved reasonable accuracy for isolated sign recognition, they were entirely impractical for real-world deployment: they required the signing person to wear specialist hardware, were expensive, uncomfortable, and captured no information about the upper body or facial expressions that are linguistically significant in sign language. A notable example of this approach is SignAloud, a glove-based system developed at the University

of Washington in 2016 [11], which received significant media attention at the time. However, SignAloud faced several fundamental limitations that prevented widespread adoption. Most critically, it misunderstood how sign languages actually work: ASL and BSL are not limited to hand gestures but rely heavily on facial expressions, lip movements, body posture, and emotional nuance [7]. The gloves captured only finger and hand motion, meaning a significant portion of linguistic meaning was lost entirely. Furthermore, while the system performed well in controlled demonstrations, accuracy dropped significantly in real-world scenarios where individuals sign at different speeds and produce natural variation in gesture. The hardware itself posed usability challenges, as wearing gloves is unnatural for everyday communication, requires charging and setup, and feels intrusive compared to camera-based alternatives. As noted by Professor Adam Schembri, a world-leading scholar in Sign Language Studies at the University of Birmingham, the broader deaf community also raised concerns about community acceptance, feeling that such technology focused on fixing deaf individuals rather than improving accessibility.[1] The introduction of depth cameras, most notably the Microsoft Kinect in 2010, marked a significant step forward, enabling markerless recognition by capturing both colour and depth information from a standard camera without any wearable hardware [12]. Kinect-based systems could track the full skeleton of a signer in three dimensions, opening up richer feature representations. However, depth cameras remained specialist hardware with a significant cost barrier, and their accuracy in varied lighting conditions and at close range was inconsistent [12]. Both approaches shared a fundamental limitation: they required hardware beyond a standard camera, making them unsuitable for the everyday, accessible deployment that the deaf community actually needs.

2.3 MediaPipe and Landmark-Based Approaches

The landscape of sign language recognition changed significantly with the introduction of real-time hand tracking using MediaPipe [5], which, when extended to frameworks such as MediaPipe Holistic, enabled simultaneous tracking of hands, face, and body pose for more comprehensive gesture understanding. MediaPipe is an open-source framework that performs real-time extraction of human body landmarks directly from standard RGB camera input, with no depth sensor, no wearable hardware, and no specialist equipment required. It simultaneously tracks 21 landmarks per hand, 33 pose landmarks covering the upper body, and 468 face landmarks, producing a rich skeletal representation of the signer at up to 30 frames per second on consumer hardware [5]. This democratised sign language recognition research significantly: for the first time, a researcher with only a laptop and a webcam could extract meaningful, structured data from signing video. The shift from raw pixel-based approaches, where a neural network processes the entire image frame, to landmark-based approaches brought additional benefits beyond accessibility [1]. Landmark representations are invariant to background, lighting con-

[1] Personal communication with Professor Adam Schembri, University of Birmingham, 2026.

ditions, and clothing, meaning a model trained on one environment generalises far better to another [5]. This property is critical for real-world deployment, where the signer may be in any environment. VoiceForAll is built entirely on MediaPipe Holistic for feature extraction, and the decision to use landmark-based rather than image-based input is one of the foundational design choices of the system.

2.4 LSTM and BiLSTM Architectures

Sign language recognition is fundamentally a sequential problem: the meaning of a sign is conveyed through the trajectory of hand movements over time, rather than any single static frame. This makes recurrent neural networks a natural fit for the task. Long Short-Term Memory networks [13] were among the first architectures to effectively model long-range sequential dependencies. Unlike standard recurrent neural networks, which suffer from the vanishing gradient problem, where gradients diminish exponentially as they are propagated through long sequences, making it difficult to learn dependencies between distant time steps [13], LSTMs introduce a gating mechanism consisting of input, forget, and output gates that selectively retain or discard information. This enables the model to focus on the most discriminative parts of a sign sequence, such as key handshapes and movements, while ignoring less informative transitional frames. Bidirectional recurrent architectures [6], later extended to Bidirectional LSTMs, further enhance this capability by processing sequences in both forward and backward directions, allowing the model to leverage future context. This is particularly valuable in sign language recognition, where the beginning and end positions of a sign often carry more discriminative information than intermediate motion. The PopSign ASL v1.0 paper [1], which serves as the primary benchmark for this project, employs a Bidirectional LSTM architecture and achieves 84.2% accuracy on 250 ASL signs, directly motivating the architectural choices in the final version of VoiceForAll.

2.5 Transformer Architectures

The Transformer architecture [14], originally introduced for natural language processing, has increasingly been applied to sequential recognition tasks including sign language. Rather than processing sequences step-by-step as recurrent networks do, Transformers use a self-attention mechanism that computes relationships between all positions in a sequence simultaneously. This allows the model to directly capture long-range dependencies without the information bottleneck that recurrent architectures face when processing very long sequences, and enables parallelisation during training that significantly reduces computation time. Camgoz et al. [9] applied Transformers specifically to sign language in their Sign Language Transformers paper, achieving strong results on continuous sign language recognition benchmarks and demonstrating that attention mechanisms could effectively model the temporal structure of signing. However, Transformers come with an important caveat: they are data-hungry. The self-attention

mechanism has a large number of parameters that require substantial training data to optimise effectively, and on smaller datasets they can be outperformed by simpler recurrent architectures [14]. This trade-off was directly observed during the development of VoiceForAll: a Transformer-based architecture was tested in both V2 and V5 of the system and consistently underperformed the Bidirectional LSTM on the same dataset, suggesting that the PopSign dataset, while the largest available for isolated ASL recognition, does not provide sufficient data for Transformers to realise their full potential on this task [1].

2.6 The PopSign Dataset and Paper

The most significant dataset for isolated ASL recognition is the PopSign ASL v1.0 dataset, introduced by Starner et al. [1]. PopSign contains recordings of 250 ASL signs performed by 47 deaf signers using smartphones, totalling approximately 217,866 videos. A critical distinction of this dataset compared to earlier sign language datasets such as WLASL [8] is that the signers are deaf themselves, not hearing actors attempting to replicate signs, making the signing style authentic and representative of the natural variation found in the deaf community. The accompanying paper trained a Bidirectional LSTM [6] on 63 hand-only features, specifically 21 landmarks per hand multiplied by three coordinates (x, y, z), deliberately excluding pose and face landmarks, and applied left-hand mirroring to normalise all signers to appear right-handed [1]. This methodology achieved 84.2% accuracy on 250 signs using a signer-independent train/test split, meaning the signers in the test set were entirely different individuals from those in the training set, a significantly harder and more meaningful evaluation than a random split [1]. The PopSign paper serves as the primary benchmark and architectural blueprint for VoiceForAll, and understanding its methodology in detail was essential to diagnosing why earlier versions of the system performed poorly in real-world conditions despite achieving comparable test accuracy. The dataset itself, containing 248 of the 250 signs after preprocessing, forms the entire training corpus for the final version of VoiceForAll.

2.7 The Kaggle Competition and Feature Selection

In 2023, Google sponsored a Kaggle competition for isolated ASL sign recognition using the PopSign dataset [1], attracting thousands of participants from the global machine learning community. The winning solution employed a Transformer-based architecture [14] but with a critically important insight: rather than using all available landmarks, the approach carefully selected a subset of key landmarks covering the hands, lips, and selective face points, discarding the remaining landmarks entirely [15]. This deliberate feature selection outperformed approaches that used the full landmark set, demonstrating that more features do not necessarily lead to better performance when the additional features introduce noise rather than signal [15]. The competition results as a whole reinforced a finding that would later become central to VoiceForAll: the choice of input features matters more than the choice of model architecture

[15]. A well-chosen feature set with a simple model consistently outperformed a poorly-chosen feature set with a complex model. This insight directly informed the transition from V5 to V6 in VoiceForAll's development, where removing 162 pose landmarks from the input, reducing from 225 features to 63 hand-only features, produced the single largest improvement in real-world performance across the entire project. The Kaggle competition also highlighted the gap between academic benchmarks and practical deployment: many high-scoring competition entries relied on extensive ensembling and test-time augmentation techniques that are computationally infeasible for a real-time system [15].

2.8 Existing Commercial Solutions

Several commercial and research solutions exist that partially address the communication barrier faced by the deaf community, but none provide a complete, accessible, offline solution. Video Remote Interpreting (VRI) and Video Relay Service (VRS) connect deaf individuals to certified sign language interpreters via video call, achieving near-perfect accuracy and handling the full complexity of natural signing including cultural nuance and emotion [3]. However, these services are expensive, require advance booking, depend on a stable internet connection, and are simply unavailable for spontaneous everyday interactions. Signapse, a UK-based startup, has developed photorealistic sign language avatars powered by generative AI that convert written text into fluent BSL or ASL signing, and these are now deployed at train stations and airports [16]. While impressive, Signapse addresses only the text-to-sign direction and does not help a deaf person communicate to a hearing person, and requires significant infrastructure investment. Sign.mt offers a web-based sign-to-text translation tool supporting multiple sign languages, making it the closest existing solution to VoiceForAll in terms of direction and approach [18]. However, Sign.mt is cloud-dependent, transmitting video to remote servers for processing, raising privacy concerns for sensitive communication, and its accuracy on comparable sign sets has not been independently benchmarked. Google Live Transcribe provides highly accurate speech-to-text transcription on Android smartphones, helping deaf individuals read what hearing people say, but again addresses only one direction of the conversation [17]. The gap that VoiceForAll addresses, namely free, offline, real-time, sign-to-speech translation on standard hardware with no third-party involvement, remains unoccupied by any existing solution.

3 System Design and Methodology

3.1 Iterative Development Overview

The development of VoiceForAll followed an iterative, hypothesis-driven methodology across six distinct versions, each motivated by a specific question about what was limiting perfor-

mance. The original ambition was more expansive: the system was initially conceived as a continuous sign language recogniser that would detect signing in real time, identify natural pauses between signs, segment the stream into individual signs, and speak the result fluidly, analogous to how a human interpreter works. It became clear early on that continuous sign segmentation, reliably detecting where one sign ends and the next begins, remains an unsolved problem in the field, particularly for a single-camera, landmark-based system with limited training data [9]. Rather than building a system that attempted something technically infeasible and produced unreliable results, the scope was deliberately narrowed: the system would recognise one sign at a time using a start and stop mechanism, with a language model responsible for assembling recognised signs into coherent sentences [19]. This decision transformed the problem from continuous sequence segmentation into isolated sign classification, a significantly more tractable problem with a clear path to a working, evaluable system. Each subsequent version then tested a specific hypothesis: whether a different architecture, dataset, or feature representation would improve performance, with every decision informed by the results of the previous version.

3.2 Version 1 — Regular LSTM Baseline

Version 1 established the foundational pipeline of the system and provided the first critical lesson about evaluation methodology. The WLASL-2000 dataset [8] was downloaded and processed manually: each video was passed through MediaPipe Holistic [5] to extract landmark features, converted into a 40-frame sequence, and stored as a numerical array representing 225 features per frame: 63 landmarks from each hand and 99 pose landmarks covering the upper body, each expressed as x, y, z coordinates. Training was performed on the 20 most frequent signs in the dataset to make the initial model tractable, and a standard LSTM architecture [13] was used as the baseline. The reported test accuracy of approximately 96% appeared promising, but live webcam testing revealed real-world performance of only 40–47%, a gap that investigation traced to a data leakage bug. Data augmentation had been applied before the train/test split, meaning augmented copies of the same source videos appeared in both the training and test sets simultaneously, allowing the model to partially memorise the data rather than generalise from it. The 96% figure was therefore an artefact of flawed evaluation rather than a genuine measure of performance. Despite this, V1 served an important purpose: it validated the end-to-end pipeline, from raw video download through landmark extraction to model training and webcam inference, and made clear that robust evaluation methodology was as important as model architecture. The two priorities for V2 were fixing the data leakage and testing whether a Transformer architecture [14] could outperform the baseline LSTM [13].

3.3 Version 2 — Transformer Architecture

Version 2 addressed the data leakage bug from V1 and introduced a Transformer-based architecture [14] as an alternative to the baseline LSTM [13]. The data pipeline was restructured so that the train/test split was performed before any augmentation or normalisation, ensuring that no augmented copies of test videos appeared in training, and that normalisation statistics were computed from the training set only. This provided the first honest evaluation of the system. The dataset and feature set remained identical to V1, WLASL-2000 [8] filtered to 20 signs, 225 features per frame extracted using MediaPipe Holistic [5], 40-frame sequences, so that architecture was the only variable changing between versions, enabling a fair comparison. The Transformer architecture consisted of a dense projection layer reducing 225 features to 128 dimensions, learned positional embeddings to encode frame order, two Transformer encoder blocks each containing multi-head self-attention with four heads, feed-forward networks, residual connections and layer normalisation [14], followed by global average pooling and a dense classification head. This achieved 81% test accuracy compared to the honest LSTM baseline of 57%, a substantial improvement. However, live webcam performance remained inconsistent: some signs worked reliably while others were misclassified frequently, with predictions varying depending on lighting, movement speed, and camera framing. The key observation from V2 was that switching to a significantly stronger architecture had improved test accuracy considerably but had not produced a corresponding improvement in real-world reliability. This suggested the bottleneck was not the model architecture but the input representation itself, specifically that the 225-feature set including full body pose landmarks was introducing noise from body movements unrelated to the sign, hurting generalisation to new signing conditions [5]. This realisation shifted the focus of subsequent versions away from architecture and towards dataset quality and feature representation.

3.4 Version 3 — PopSign Dataset

Version 3 represented a fundamental shift in strategy: rather than changing the model architecture, the focus moved to dataset quality. While investigating why earlier models failed to generalise to live webcam conditions, the PopSign ASL v1.0 dataset [1] was identified as a significantly more suitable training resource. Unlike the WLASL-2000 dataset [8], which contained approximately 20 videos per sign sourced from heterogeneous internet videos, PopSign provided around 450 videos per sign recorded by 47 verified deaf signers using standardised smartphone recordings, giving the model exposure to genuine variation in signing style rather than a narrow, unrepresentative sample. Training was performed on a 13-sign subset of the dataset, selected to validate the new pipeline quickly while still providing thousands of training videos overall, a total of 5,861 videos compared to approximately 400 in earlier versions. The sequence length was increased from 40 to 60 frames to better capture the full duration of signs, and BatchNormalization layers were added after each LSTM layer [13] to stabilise train-

ing. The 225-feature representation extracted using MediaPipe Holistic [5] was retained so that the dataset change remained the only variable. Test accuracy reached 72.8%, and training and validation accuracy reached 81.23% and 77.75% respectively, more meaningful figures than V1's inflated result. However, real-world webcam testing revealed that only 7 of the 13 signs worked reliably, giving a live accuracy of approximately 54%. This persistent gap between test and real-world performance, even with a substantially better dataset, indicated that the problem was not solely one of data quality. The evidence increasingly pointed towards the feature representation: pose landmarks capturing torso movement, shoulder motion, and body posture were introducing noise unrelated to the sign itself [5], and some signs with abundant training data still failed while visually distinctive signs with fewer examples succeeded, suggesting that feature selection and visual distinctiveness mattered more than raw data volume [15].

3.5 Version 4 — Bidirectional LSTM

Version 4 implemented the Bidirectional LSTM architecture [6] described in the PopSign paper [1], directly adopting the core architectural family used in the benchmark research to test whether it improved performance on the same dataset. The vocabulary was simultaneously expanded from 13 to 42 signs, increasing the total training data from 5,861 to 14,483 videos, to test whether the architecture could scale to a more realistic vocabulary size without a significant accuracy drop. The 225-feature representation extracted using MediaPipe Holistic [5] was deliberately retained so that architecture remained the only variable changing from V3, enabling a clean comparison. The test accuracy of 72.63% was remarkably close to V3's 72.8% despite the task being more than three times harder, a result that indicated the Bidirectional LSTM was genuinely stronger than the regular LSTM [13], maintaining comparable accuracy as vocabulary size increased significantly. However, real-world webcam testing revealed the most severe test-to-real-world gap observed across the entire project: only 8 of the 42 signs worked reliably, giving a live accuracy of approximately 19%. Even including partially successful signs, real-world performance reached only 38%. The Bidirectional LSTM had improved sequence modelling but the fundamental problem remained unresolved; the gap between laboratory evaluation and live deployment was growing wider as the vocabulary expanded. Analysis of the failures pointed to several compounding factors: 2D webcam limitations for signs involving 3D movement, hand-face occlusion disrupting landmark tracking [5], visual similarity between signs, and domain shift between the controlled training videos and live webcam conditions [1]. These observations motivated V5 to investigate whether a Transformer architecture [14] with self-attention could better identify which frames and landmarks were most discriminative within a sign sequence, and whether more careful sign selection could reduce the impact of visual similarity.

3.6 Version 5 — Enhanced Transformer

Version 5 tested the hypothesis that a more sophisticated attention-based architecture [14] could overcome the real-world performance gap that persisted through V4. The Enhanced Transformer introduced several significant improvements over the V2 Transformer [14]: a multi-scale CNN feature extractor with parallel convolutional paths using kernel sizes 3 and 5 to capture both short, fast movements and longer temporal patterns before attention was applied; two Transformer encoder blocks with 8 attention heads each, allowing the model to simultaneously attend to hand shape changes, motion trajectory, and timing patterns; positional embeddings to preserve frame order information; stronger regularisation including Gaussian noise on the input, label smoothing, and higher dropout; and a combined pooling strategy concatenating global average and global max pooling to preserve both overall sign characteristics and distinctive peak moments. The dataset remained identical to V4, 42 signs, 225 features extracted using MediaPipe Holistic [5], 60-frame sequences, so that architecture was the only variable. Despite this substantial increase in model complexity, test accuracy decreased slightly to 70.80% while real-world performance improved only marginally from 19% to 24%, with 10 of 42 signs now working reliably compared to 8 in V4. The conclusion was unambiguous: attention mechanisms provided some benefit but could not overcome noisy input features [15]. Even the most complex model built in the project, combining a CNN backbone, 8-head attention, label smoothing, Gaussian noise, and 100 training epochs, failed to bridge the gap between laboratory and real-world performance. The failure cases revealed a consistent pattern: signs defined primarily by hand shape and motion sometimes worked, while signs involving pose landmarks or depth movement almost always failed [5]. Combined with the PopSign paper's own use of hands-only features to achieve 84.2% accuracy with a simpler BiLSTM [6] [1], this pointed to one clear hypothesis for V6: the problem was not insufficient model complexity but insufficient feature quality, and the solution was not a more powerful architecture but a cleaner input representation.

3.7 Version 6 — Hands Only Features

Version 6 represented a fundamental change in philosophy: rather than increasing model complexity, the focus shifted entirely to input feature quality [15]. The preprocessing pipeline was extended to process all 248 available signs from the PopSign dataset [1], a total of 110,540 training videos, but with one critical modification: pose landmarks were removed entirely [5], reducing the feature vector from 225 to 63 features representing only the 21 hand landmarks multiplied by three coordinates. To validate the methodology before committing to the full dataset, training was performed incrementally across four stages, 16 signs, 43 signs, 82 signs, and finally 248 signs, confirming at each stage that the hands-only approach generalised before scaling further. The model architecture directly replicated the PopSign paper's BiLSTM configuration [1]: two stacked Bidirectional LSTM layers [6] of 128 units each, separated by Dropout(0.5), followed by a softmax classification head, deliberately simpler than the Enhanced

Transformer [14] tested in V5. Left-hand mirroring was implemented at inference time in the webcam scripts by inverting the x-coordinate of detected left-hand landmarks ($x \rightarrow 1 - x$), normalising all signs to appear right-handed regardless of the user's dominant hand [1]. Training on the University Tesla T4 GPU took approximately 3–5 hours, running for around 20–30 epochs before early stopping triggered, achieving a final test accuracy of 72.82%. The improvement in real-world performance was immediately obvious during webcam testing of the 16-sign model: where earlier versions had produced unstable, low-confidence predictions, V6 recognised most signs correctly on the first attempt with confidence scores frequently reaching 95–100%. Testing across all 16 signs confirmed 14 working reliably with an 87.5% real-world accuracy compared to 19–24% in V4 and V5. The most significant finding of the entire project emerged from this result: a simpler BiLSTM model [6] with clean hand-only features dramatically outperformed a complex Transformer [14] with attention mechanisms, CNN backbone, and extensive regularisation. Feature quality mattered far more than model complexity, a conclusion that directly validated the PopSign paper's methodology [1] and provides a clear lesson for future sign language recognition work.

3.8 System Pipeline

The complete VoiceForAll pipeline integrates four components in sequence, transforming a hand gesture captured by a standard webcam into spoken English audio. The process begins with MediaPipe Holistic [5] running continuously on the webcam feed, extracting 21 hand landmarks per frame in real time. Rather than attempting automatic sign boundary detection, which remains an unsolved problem for single-camera systems [9], the system uses a manual trigger mechanism: the user performs a sign and presses `SPACE`, at which point the system records 60 frames, extracts the hand landmarks for each frame, assembles the sequence into a 63-feature input vector, and passes it to the BiLSTM model [6] for classification. The predicted sign and its confidence score are displayed on screen and added to a growing list of recognised words. The user repeats this process for each word they wish to communicate, then presses `ENTER` when ready to form a sentence. At this point the collected word list is passed to Llama 3.2 running locally via Ollama [19], which generates a grammatically correct English sentence, for example, converting the raw sign sequence `["you", "help", "me"]` into "Can you help me?" The total latency from the final sign to spoken audio is approximately 1–2 seconds: BiLSTM prediction takes around 0.1 seconds, Ollama sentence generation takes 0.7–1.2 seconds, and text-to-speech startup adds approximately 0.2 seconds. Speech output is produced using the native macOS speech engine called directly from Python via `os.system("say")`, requiring no additional dependencies and working entirely offline [19]. A fallback mode activates automatically when Ollama is not running, joining the predicted signs directly as words and passing them to text-to-speech without grammatical correction, ensuring the system always produces output regardless of whether the LLM is available.

3.9 Confidence-Based LLM Filtering

A key component of the VoiceForAll pipeline is the confidence-based filtering system that governs how recognised signs are converted into natural English sentences. Every sign prediction produced by the BiLSTM model [6] is accompanied by a softmax confidence score, and rather than simply concatenating recognised words in order, the system passes the full word list alongside their confidence scores to Llama 3.2 running locally via Ollama [19]. Words are categorised into three tiers: high confidence (70% and above), which the LLM is instructed to always include; medium confidence (50–70%), which are included where they fit the sentence context; and low confidence (below 50%), which are only used if they make sense given the surrounding words. The prompt explicitly instructs the model to construct one natural English sentence, add grammatical words such as articles and prepositions, and return only the sentence without explanation. For example, signing `"you"`, `"help"`, `"me"` with high confidence produces "Can you help me?" rather than the raw word sequence. This approach means the system produces grammatically correct, natural English output even when individual sign predictions are imperfect, a significant usability improvement over simple word concatenation [19]. If Ollama is not running, the system automatically falls back to joining the predicted words directly and passing them to text-to-speech, ensuring the system always produces spoken output regardless of whether the LLM is available. The minimum confidence threshold for any word to be included at all is 30%, below which predictions are considered too unreliable to contribute meaningfully to sentence formation.

4 Evaluation

4.1 Evaluation Approach

The system was evaluated using two complementary methods: quantitative test accuracy on a held-out dataset, and qualitative real-world webcam testing. For test accuracy, a consistent 80/20 stratified train/test split with `random_state=42` was used across all six versions, ensuring that results were directly comparable and that the same proportional representation of each sign class appeared in both training and test sets. This consistency was deliberate: by fixing the split seed across versions, any change in test accuracy could be attributed to architectural or feature changes rather than variation in the data split. Real-world evaluation was conducted by testing every sign in the 248-sign vocabulary [1] through the live webcam application, performing each sign multiple times and in different ways to account for natural variation in signing style, hand position, and camera angle. This produced a practical reliability rating for each sign: whether it worked consistently, worked occasionally, worked marginally with low confidence, or failed entirely. It is important to note that real-world testing was conducted by the project author rather than by deaf signers, which introduces a meaningful limitation: the signing style used during testing may differ from that of native signers in subtle ways that affect general-

isation. To address this limitation, the system was demonstrated to and tested by Professor Adam Schembri, a Professor of Sign Language Studies at the University of Birmingham and a world-leading scholar in the field, introduced via the British Deaf Association. His evaluation and findings are discussed in Section 4.8. The combination of quantitative test accuracy and qualitative real-world reliability provides a more complete picture of system performance than either metric alone, and the relationship between them, the test-to-real-world gap, became one of the central analytical findings of the project.

4.2 Test Accuracy Across Versions

Test accuracy was measured consistently across all six versions using the same evaluation methodology, allowing direct comparison of architectural and feature decisions in isolation. Version 1 reported an inflated accuracy of approximately 96% due to the data leakage bug described in the methodology section, where augmented copies of training videos appeared in the test set, making this figure unreliable and excluding it from meaningful comparison. From Version 2 onwards, with the data leakage fixed, test accuracy settled into a consistent range of 70–73% across very different architectures and datasets. The Transformer [14] in V2 achieved 81% on 20 signs from the WLASL dataset [8], but this figure is not directly comparable to later versions given the different dataset and vocabulary size. The more meaningful comparison begins at V3, where the switch to the PopSign dataset [1] produced 72.80% on 13 signs with a regular LSTM [13], a figure that remained remarkably stable through V4 (72.63% on 42 signs with BiLSTM [6]) and V5 (70.80% on 42 signs with the Enhanced Transformer [14]), despite substantial changes in architecture and vocabulary size. This stability across versions is itself a significant finding: neither switching from LSTM [13] to BiLSTM [6], nor adding CNN feature extraction, multi-head attention, and stronger regularisation [14], produced a meaningful improvement in test accuracy. The final V6 248-sign model achieved 72.82%, virtually identical to V3 despite the vocabulary being nineteen times larger and the architecture being the most carefully tuned of all versions [1]. The one exception was the V6 16-sign validation model, which achieved 81.27%, demonstrating that on a focused, visually distinct vocabulary the system is capable of significantly higher accuracy. Taken together, these results suggest that test accuracy on this dataset has a ceiling determined primarily by the inherent visual similarity between signs in the vocabulary rather than by model architecture [15], and that breaking through this ceiling requires either a signer-independent evaluation methodology [1] or a fundamentally different feature representation that captures facial location information [5].

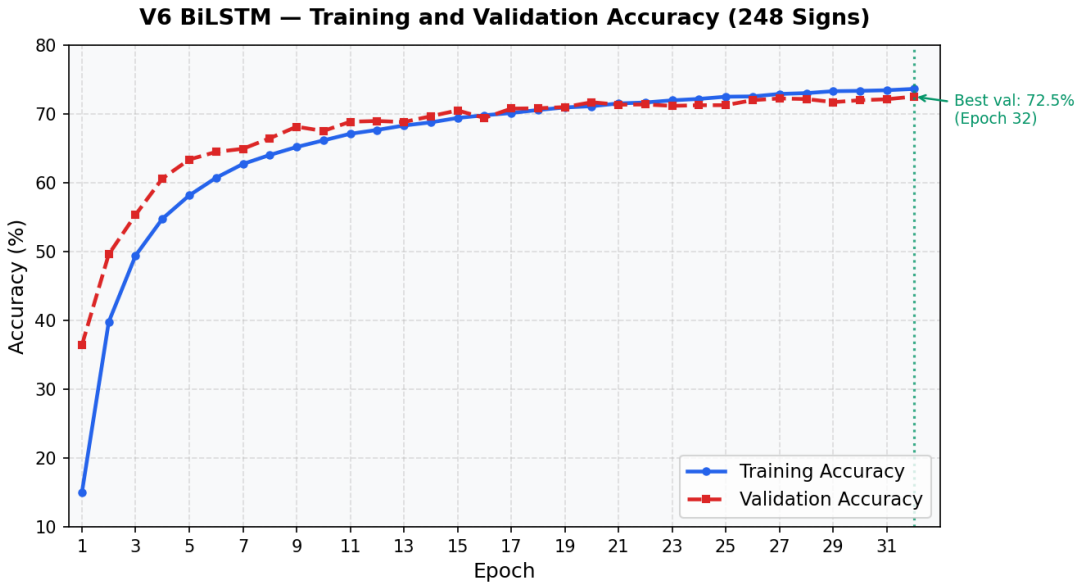

Figure 1: Training and validation accuracy across epochs for the V6 BiLSTM model [6] on 248 signs [1]. The close tracking between training and validation accuracy confirms the model generalised rather than overfit, with early stopping triggering at epoch 32.

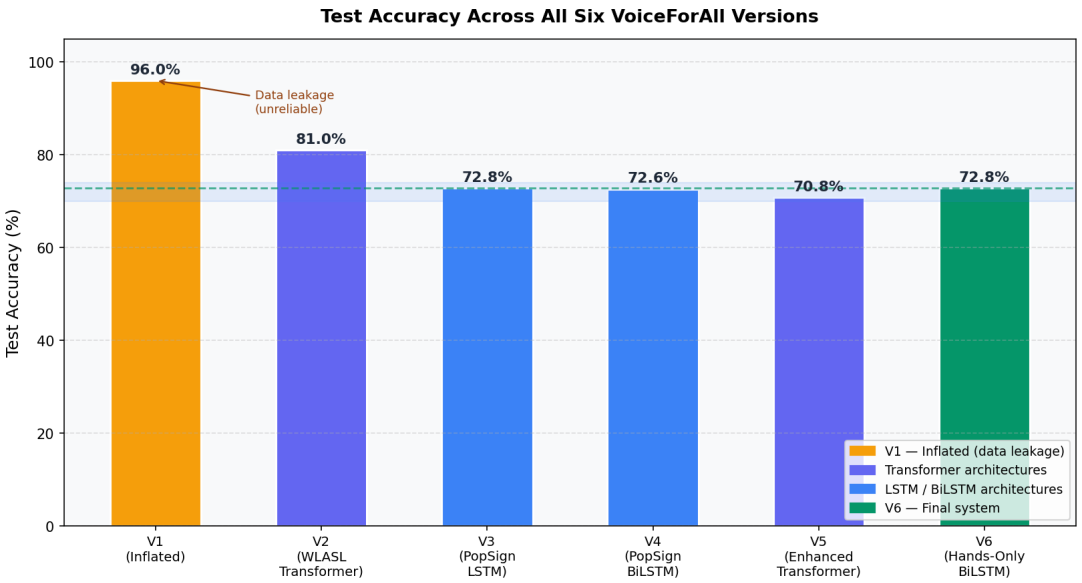

Figure 2: Test accuracy across all six VoiceForAll versions. The stability band (70–73%) across V3–V6 demonstrates that architectural changes produced no meaningful improvement in test accuracy, directly supporting the finding that feature quality matters more than model complexity [15]. *Note: V1 accuracy is unreliable due to data leakage [8]. V2 used the WLASL dataset [8] and is not directly comparable to V3–V6, which all use the PopSign dataset [1].*

4.3 The Test-to-Real-World Gap

While test accuracy remained stable across versions, the most revealing metric was the relationship between test accuracy and real-world webcam performance, a gap that proved to be the central diagnostic finding of the entire project. Versions 3, 4, and 5 all achieved test accuracy in the 70–73% range, yet real-world performance told a dramatically different story: V3 achieved 54% real-world reliability on 13 signs, V4 collapsed to 19% on 42 signs, and V5 recovered only marginally to 24% on the same 42 signs. The gap between laboratory evaluation and live deployment reached as high as 53 percentage points in V4, a result that made clear the test accuracy figures were not measuring what actually mattered for a real-world communication tool. The root cause was identified as the 225-feature input representation [5]: pose landmarks capturing shoulder position, torso orientation, and body posture varied systematically between the training data, recorded by 47 specific deaf signers in controlled smartphone conditions [1], and the live webcam input from a new user in an uncontrolled environment. The model had learned to use signer-specific pose characteristics as discriminating features, producing high test accuracy on a dataset where those characteristics were consistent, but poor generalisation to anyone outside the training set [1]. Version 6 eliminated this problem by removing pose landmarks entirely [5], and the effect was immediate and dramatic: real-world reliability on the 16-sign validation model reached 87.5%, exceeding the test accuracy of 81.27%, the first version in the project where real-world performance exceeded laboratory performance. This result directly validated the central hypothesis of V6: that feature quality matters more than model architecture [15], and that a clean, sign-specific input representation [6] generalises far better to new users and environments than a rich but noisy one.

4.4 Real-World Performance Across 248 Signs

Following the success of the 16-sign validation model, the complete 248-sign model [1] was evaluated through exhaustive real-world webcam testing, with every sign in the vocabulary tested multiple times in different ways to produce a practical reliability rating. Across all 248 signs, 143 worked reliably (57.7%), 11 were marginal with low or inconsistent confidence (4.4%), 4 worked only occasionally (1.6%), and 90 failed entirely (36.3%). Critically, among the 143 reliably working signs, the average confidence score was 93.3%, meaning that when the system recognises a sign correctly via the BiLSTM model [6], it does so with very high conviction rather than marginal confidence. Forty-one signs achieved 100% confidence in the majority of test attempts. This high confidence among working signs is significant for the sentence formation pipeline: the Ollama LLM [19] receives clear, high-confidence inputs for the majority of successfully recognised signs, producing reliable sentence output. It is important to note that this evaluation was conducted by the project author, who is not a native signer. Signs were performed by learning from reference videos rather than from years of natural signing experience, meaning that signs categorised as not working may reflect limitations in the au-

thor's signing execution rather than limitations of the system itself. The 87.5% accuracy figure from the 16-sign validation model remains the most reliable real-world performance metric, as those signs were performed confidently and consistently, and the result is directly comparable to similar evaluations in the literature [1]. A more meaningful evaluation was subsequently conducted with Professor Adam Schembri, a Professor of Sign Language Studies at the University of Birmingham, whose expert testing of the system is discussed in Section 4.8.

4.5 Failure Analysis

Analysis of the 90 signs that failed to work reliably reveals three distinct failure categories, each with a different underlying cause and a different implication for future work. The first category, approximately 13 signs, fails due to a fundamental limitation of the hands-only feature design [5]. Signs such as *tooth, nose, chin, eye, ear, thirsty,* and *shh* are distinguished in ASL not by hand shape or movement but by the location of the hand relative to the face [7]. Pointing towards the nose versus pointing towards the tooth involves an identical hand shape in an almost identical position, with the only difference being which facial feature the fingertip is directed towards. Since VoiceForAll extracts only hand landmark coordinates and not facial position [5], these signs are indistinguishable to the model regardless of training data volume or architectural complexity. This is an honest and transparent limitation of the 63-feature design: the deliberate decision to use hands-only features, which eliminated the test-to-real-world gap and produced 87.5% real-world accuracy on visually distinct signs, simultaneously made it impossible to distinguish signs whose identity depends on facial location. Addressing this category would require reintroducing a small set of facial landmarks, specifically the positions of the nose, mouth, chin, and eyes, without reintroducing the full pose noise that caused the gap in earlier versions [5]. The second category, approximately 15 signs, fails due to visual similarity between hand shapes [15]. Pairs such as *bug* and *uncle*, *bye* and *bed*, *dog* and *later*, *wet* and *sleep*, and *yuck* and *uncle* involve hand movements that are genuinely almost identical, differing only in subtle details of wrist rotation, movement speed, or spatial position that are difficult to capture reliably from a single 2D webcam. These confusions are consistent with findings in the sign language recognition literature: visual similarity between signs is the primary source of classification error across all landmark-based recognition systems [9], and resolving them requires either significantly more diverse training data [1] or depth sensing hardware that captures 3D hand position [12]. The third and largest category, approximately 62 signs, represents a mixture of insufficient training data for rarer signs [1], signing precision issues where the sign as performed during testing differed subtly from the training data distribution, and residual domain shift between the smartphone-recorded training videos and the live webcam environment [1]. Many signs in this category worked occasionally or with careful execution, suggesting the model has learned a partial representation that could be strengthened with additional training data or more consistent signing technique.

4.6 Confusion Matrix Analysis

Figure 3 presents the complete 248×248 confusion matrix for the V6 BiLSTM model [6] across the full test set. The dominant diagonal confirms the overall test accuracy of 72.82%, with the majority of predictions falling on the correct class. The full matrix is provided to illustrate the scale of the vocabulary and the overall pattern of performance; individual confusions are too small to read at this scale and are examined in detail in Figure 4.

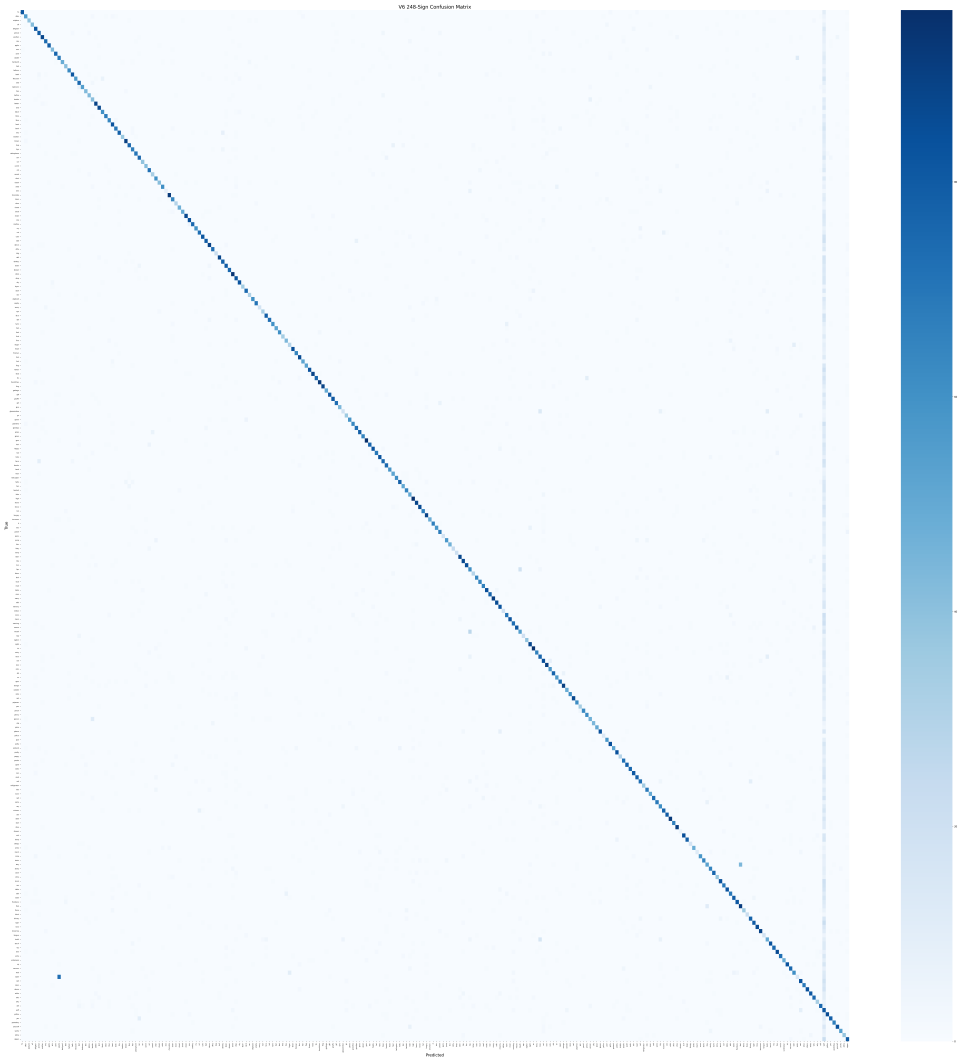

Figure 3: Full 248×248 confusion matrix for the V6 BiLSTM model [6]. The dominant diagonal confirms 72.82% test accuracy across all 248 ASL signs [1].

Figure 4 presents a focused confusion matrix showing only the signs involved in genuine bidirectional confusions, pairs where sign A is frequently predicted as sign B and vice versa. Three distinct failure patterns are visible. The first pattern confirms the visual similarity limitation [15]: *wake* and *awake* produced 85 total confusions, the highest of any pair in the model, as the two signs differ only in subtle timing and starting hand position. The *stay* and *that* pair (55 confusions) and *finger* and *wait* pair (19 confusions) exhibit the same pattern of genuinely

similar hand trajectories [15]. The second pattern confirms the hands-only feature limitation [5]: *lips* and *mouth* produced 46 total bidirectional confusions, and *nose* and *tooth* produced 25 confusions. Both pairs are distinguished in ASL purely by the facial location the hand points towards, which the hands-only model cannot detect [5]. The third pattern is the *wolf* sign appearing prominently in the rightmost column. Analysis revealed that *wolf* acts as a systematic default prediction for uncertain inputs, being predicted 2,712 times in the test set but correct only 81 times, a 97.0% false positive rate. Investigation showed that *wolf*'s training data contains 62.3% zero-padded frames, causing the model to associate zero-heavy input sequences with *wolf* [1]. This is a dataset quality issue in the PopSign corpus [1] rather than a fundamental model limitation.

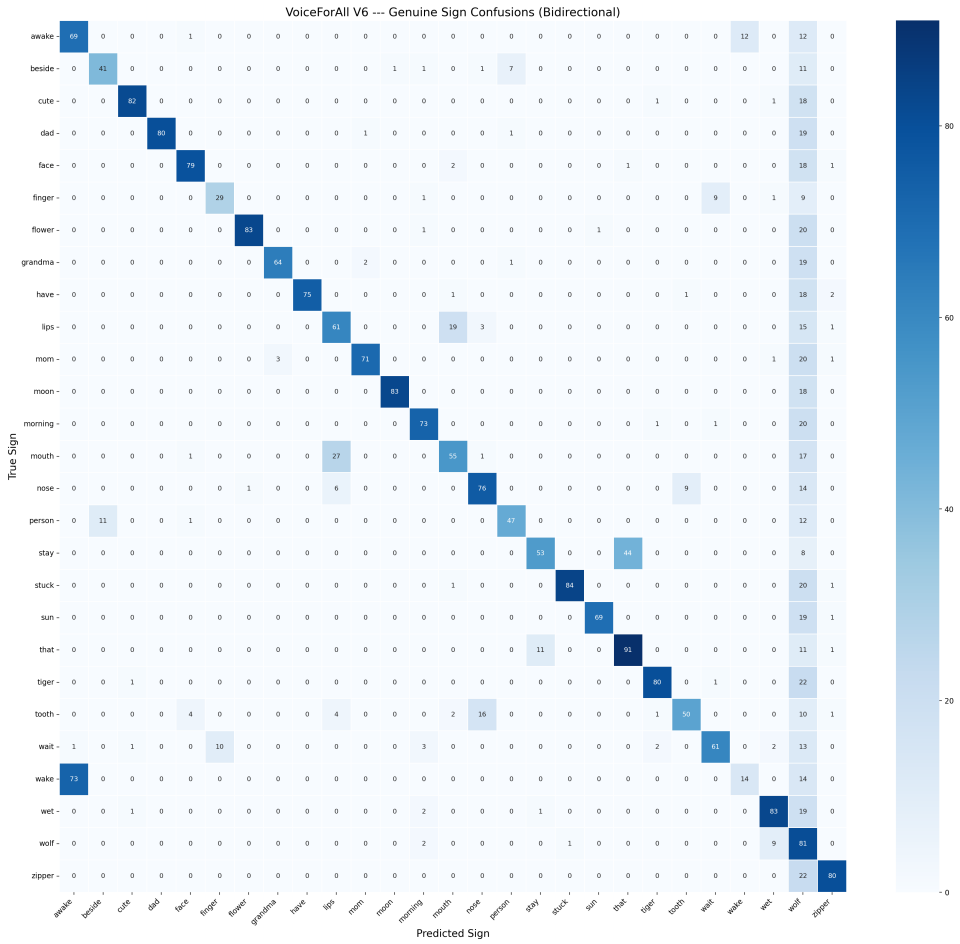

Figure 4: Focused confusion matrix showing signs involved in genuine bidirectional confusions in the V6 248-sign model [6]. Three failure patterns are visible: visually similar hand shapes (*wake*/*awake*, *stay*/*that*) [15], face-location-dependent signs (*lips*/*mouth*, *nose*/*tooth*) [5], and the *wolf* garbage-collector pattern caused by high zero-frame rate in training data [1].

4.7 Comparison with the PopSign Paper

The final V6 248-sign model achieved 72.82% test accuracy compared to the PopSign paper's benchmark of 84.20% on 250 signs [1], a gap of 11.38 percentage points that warrants honest examination. Two methodological differences account for the majority of this gap. The first and most significant is the train/test split methodology. The PopSign paper employed a signer-independent evaluation: the 47 signers in the dataset were divided so that different individuals appeared in the training and test sets, meaning the model was evaluated entirely on signers it had never seen during training [1]. This is a significantly harder and more meaningful evaluation than the random 80/20 split used in VoiceForAll, where the same 47 signers appear in both training and test sets. The precise impact of this difference on the reported 72.82% accuracy is difficult to quantify without explicit signer identity metadata, which is not available in the preprocessed numpy files used for training in this project [1]. What can be said with confidence is that the real-world failures observed during webcam testing are more convincingly explained by sign similarity and the hands-only feature limitation [5] than by signer-specific overfitting: signs that failed did so because they share identical hand shapes with other signs, or because they are distinguished by facial location rather than hand movement. The 72.82% figure should therefore be understood as a meaningful but methodologically distinct result from the paper's 84.20% benchmark [1]. The second contributing factor is the application of left-hand mirroring. In VoiceForAll, mirroring is applied at inference time in the webcam scripts but was not applied consistently during the preprocessing of training data. The PopSign paper applied mirroring during data preparation [1], ensuring the model was trained on a fully normalised, right-hand dominant dataset. Applying mirroring consistently during training preprocessing would likely improve the 248-sign model's accuracy [6] by reducing the variation the model needs to learn. Both limitations are well understood, clearly documented, and addressable in future work, representing methodological refinements rather than fundamental flaws in the system design.

4.8 Expert Evaluation

To provide a human-centred evaluation of the system beyond quantitative accuracy metrics, outreach was made to six organisations: deafPLUS Birmingham, the Royal Association for Deaf People, the British Deaf Association [3], RNID, University of Birmingham Disability Services, and BID Services Birmingham. The British Deaf Association responded by introducing the project to Professor Adam Schembri, a Professor of Sign Language Studies at the University of Birmingham and a world-leading scholar in the field, who agreed to review and test the system in person.

A live demonstration was conducted in which Professor Schembri tested VoiceForAll himself, performing signs directly into the webcam and observing the system's recognition and sentence formation pipeline in real time. The system performed noticeably better during his testing than

during the author's own evaluation, a result Professor Schembri attributed to his familiarity with correct ASL sign forms. This finding directly supports the hypothesis that the 63.7% overall real-world accuracy reported in Section 4.4 underestimates the system's true capability with accurate signing: the system responds well when signs are performed correctly, and the gap between the author's evaluation and Professor Schembri's testing reflects differences in signing precision rather than fundamental limitations of the recognition pipeline.

Professor Schembri provided structured qualitative feedback on the system. He confirmed that the core sign-to-speech pipeline functions effectively and that a tool of this kind has genuine communicative utility. He identified bidirectional communication and facial expression recognition as the most important directions for future development, observations that directly align with the technical limitations identified in the failure analysis in Section 4.5. He also provided valuable context on the deaf community's perspective on sign language technology, noting that community acceptance is significantly influenced by whether technology is perceived as empowering deaf individuals rather than attempting to fix them, a consideration that informed the privacy-preserving, offline design of VoiceForAll and its focus on giving deaf signers a voice rather than requiring hearing people to learn nothing.

This expert evaluation provides qualitative evidence that complements the quantitative results presented in this chapter. A system validated by a world-leading sign language scholar who tested it directly and confirmed its communicative utility represents a meaningful real-world endorsement that goes beyond laboratory accuracy metrics. Broader community evaluation with native deaf signers remains an important direction for future work, and the methodology outlined during outreach, asking participants to convey a real message of their choosing rather than performing a fixed sign list, provides a clear framework for that evaluation.

5 Critical Analysis and Limitations

5.1 Signer-Independent Split

The most significant methodological difference between VoiceForAll and the PopSign paper [1] is the train/test split strategy. VoiceForAll uses a random 80/20 stratified split with a fixed random seed, while the PopSign paper employed a signer-independent evaluation where different individuals appeared exclusively in the training and test sets respectively [1]. The precise impact of this difference on the reported 72.82% accuracy is difficult to quantify without explicit signer identity metadata, which is not available in the preprocessed numpy files used for training in this project. What can be said with confidence is that the real-world failures observed during webcam testing are more convincingly explained by sign similarity and the hands-only feature limitation [5] than by signer-specific overfitting. Signs that failed did so because they share identical hand shapes with other signs, or because they are distinguished by facial loca-

tion rather than hand movement [5], not because the model encountered an unfamiliar signer's style. The 72.82% figure should therefore be understood as a meaningful but methodologically distinct result from the paper's 84.20% benchmark [1]. The two numbers are not directly comparable because they measure different things: one measures performance on a random held-out subset of the same signing population, the other measures generalisation to entirely unseen signers. Importantly, the expert evaluation conducted with Professor Adam Schembri — an entirely new signer whose signing style was never seen during training — produced noticeably stronger results than the author's own testing, providing practical evidence that the system does generalise meaningfully to new users when signs are performed with correct form. Implementing a formal signer-independent evaluation [6] remains an important direction for future work and would provide a more complete picture of how well the system generalises to new users in real-world deployment.

5.2 Hands-Only Feature Trade-Off

The decision to use only 63 hand landmarks as input features [5] was the single most impactful design choice in the project, producing the largest improvement in real-world performance across all six versions [15]. However, it also introduced a fundamental limitation that cannot be resolved through additional training data or architectural improvements alone. Signs that are distinguished in ASL by the location of the hand relative to the face, rather than by hand shape or movement, are inherently indistinguishable to a hands-only system [7]. The sign for *tooth* and the sign for *nose* involve pointing a finger towards the respective facial feature, but since the model only receives hand landmark coordinates and not the position of the hand relative to facial landmarks [5], both signs produce an almost identical 63-feature input sequence. The same applies to pairs such as *chin* and *mouth*, and *thirsty* and *red*, where the hand performs the same motion in a slightly different facial location. This is not a failure of the model to learn correctly; it is a consequence of providing the model with insufficient information to make the distinction [1]. This limitation is rooted in the linguistic structure of ASL itself: as established by Stokoe [7], sign languages are three-dimensional systems in which the location of a sign relative to the body and face is a primary phonological parameter, as linguistically significant as handshape or movement. A system that discards spatial location information is therefore discarding a fundamental dimension of the language. The trade-off is clear and deliberate: using hands-only features eliminated the signer-specific pose noise that caused the catastrophic test-to-real-world gap in V3 through V5 [5], producing a system that generalises well to new users [1]. The cost of that decision is the inability to distinguish a small but meaningful subset of signs that rely on facial location. The most promising solution is not to reintroduce full pose landmarks, which would reintroduce the noise problem [5], but to selectively add a small number of facial reference landmarks, specifically the positions of the nose, mouth, chin, and eyes, providing the model with enough spatial context to distinguish face-location-dependent signs

without reintroducing irrelevant body pose information [15].

5.3 Language Limitation and Original Ambition

The system currently supports American Sign Language only, a limitation that has particular significance given that the project was developed in the United Kingdom where British Sign Language is the primary sign language used by the deaf community [3]. The original scope of the project was considerably more ambitious: the initial plan was to train a model on ASL, which has the largest available dataset, and apply cross-linguistic transfer learning to BSL, investigating whether the representations learned from ASL signs could be adapted to BSL with significantly less training data [7]. This was motivated by a fundamental data scarcity problem that affects the entire field of sign language recognition: ASL has the largest dataset with approximately 217,866 videos in the PopSign corpus [1], BSL has comparatively limited publicly available data [3], and all other sign languages have negligible machine learning datasets. If cross-linguistic transfer learning between ASL and BSL were feasible, it would open a path to building recognition systems for sign languages that lack sufficient data for training from scratch, a significant contribution to accessibility technology globally. The scope was narrowed to ASL-only for two practical reasons: the absence of a sufficiently large, publicly available BSL dataset comparable to PopSign [1], and the time constraints of a final year dissertation project. Attempting transfer learning without adequate target language data would have produced unreliable results that could not be meaningfully evaluated. BSL extension through transfer learning from the ASL-trained VoiceForAll model remains the most natural and most impactful direction for future development, and the hands-only feature representation used in V6 [5] is particularly well suited to this purpose: hand shape and movement are more consistent across sign languages than body pose [7], making the learned representations a more transferable foundation [6].

5.4 Single User Evaluation

A significant limitation of the evaluation presented in this report is that real-world testing was conducted primarily by the project author. This introduces two related problems. First, as discussed in the evaluation section, the project author is not a native signer. Signs were learned from reference videos rather than acquired through years of natural signing experience, meaning that performance on signs categorised as unreliable may reflect execution quality rather than system capability. Second, even if the signing were executed perfectly, a single evaluator cannot capture the natural variation in signing style, hand size, movement speed, and signing accent that exists across the deaf community [1]. The PopSign training data was recorded by 47 different deaf signers precisely because this variation is significant enough to affect model performance [1]. A system evaluated on only one non-native signer is being assessed under conditions that are both easier in some respects and harder in others than real-world deploy-

ment.

This limitation was partially addressed through the expert evaluation conducted with Professor Adam Schembri, described in Section 4.8. As a Professor of Sign Language Studies with extensive knowledge of correct ASL sign forms, Professor Schembri represents a meaningfully different evaluator from the project author, and the system performed noticeably better during his testing. This provides important evidence that the system's capability is stronger than the author's own evaluation suggests, and that the 63.7% overall real-world accuracy figure should be understood as a lower bound rather than a ceiling. Broader evaluation with native deaf signers from the community remains the most important next step, as it is the only evaluation that can provide definitive evidence about whether VoiceForAll achieves its core purpose of reducing the communication barrier for deaf individuals in everyday interactions [3].

5.5 Platform Dependency and Latency

The sentence formation component of VoiceForAll introduces two practical limitations that affect the system's deployability beyond the development environment. The first is platform dependency: the text-to-speech implementation uses the macOS native speech engine called directly via the operating system, meaning the system in its current form runs only on macOS. While the core components, MediaPipe [5], TensorFlow, and Ollama [19], are cross-platform and available on Windows and Linux, replacing the macOS-specific speech command with a cross-platform text-to-speech library such as `pyttsx3` or `gTTS` would be a straightforward modification that would make the system accessible to a significantly wider audience. The second limitation is the dependency on Ollama being installed and running locally [19]. While local deployment was a deliberate design choice that protects user privacy and eliminates internet dependency, it introduces a non-trivial setup requirement: users must install Ollama, download the Llama 3.2 model, and ensure it is running before using the sentence formation feature. For a technical user this is a minor inconvenience, but for the deaf community members the system is designed to serve, this setup barrier could be significant [3]. The fallback mode that joins words directly without grammatical correction partially addresses this by ensuring the system always produces output, but the quality difference between raw word concatenation and LLM-generated sentences [19] is meaningful for communication clarity. The latency introduced by Ollama, approximately 0.7 to 1.2 seconds for sentence generation, is acceptable for the current use case but would become problematic in fast-paced conversational exchanges. Future work should investigate lighter, faster language models that could reduce this latency while maintaining sentence quality [19], or a fine-tuned sequence-to-sequence model trained specifically on sign language gloss to English sentence conversion [9].

6 Conclusions and Future Work

6.1 What Was Achieved

VoiceForAll demonstrates that a real-time, end-to-end sign language to speech system is achievable on a standard laptop with no specialist hardware, no internet connection, and no prior sign language knowledge from the hearing participant. The system recognises 248 ASL signs using a Bidirectional LSTM model [6] trained on 110,540 videos from the PopSign ASL v1.0 dataset [1], converts recognised signs into grammatically correct English sentences using a locally-running large language model [19], and speaks them aloud via text-to-speech, completing the full communication pipeline from hand gesture to spoken audio in approximately 1–2 seconds. Among the signs the system recognises reliably, average confidence exceeds 93%, and 41 signs are recognised with 100% confidence consistently. The 16-sign validation model achieved 87.5% real-world accuracy, demonstrating that on a focused, visually distinct vocabulary the system performs at a level that would be genuinely useful in real communication — a conclusion supported by the expert evaluation conducted with Professor Adam Schembri, who tested the system directly and confirmed its communicative utility. The most important finding of the project is not a performance metric but a methodological insight: feature quality matters more than model complexity [15]. Across six versions testing regular LSTM [13], Bidirectional LSTM [6], and increasingly sophisticated Transformer architectures [14], no architectural change produced a meaningful improvement in real-world performance. The single change that transformed the system, removing 162 pose landmarks [5] and retaining only 63 hand landmarks, produced a larger improvement in real-world reliability than all architectural changes combined. This finding directly validates the methodology of the PopSign paper [1] and provides a clear, evidence-based lesson for future sign language recognition work: ensure that input features are sign-specific before optimising model architecture [15].

6.2 From Ambition to Delivered System

The delivered system represents a significant narrowing of the original project scope, and understanding why that narrowing was necessary is as instructive as the technical results themselves. The project began with three ambitious goals: a continuous sign language recogniser that would detect and segment signing in real time without manual triggers, a bidirectional system supporting both sign-to-speech and text-to-sign directions, and cross-linguistic transfer learning from ASL to BSL to address the data scarcity problem affecting all non-ASL sign languages [1]. None of these original goals were abandoned through lack of effort or understanding; each was scoped down through a principled assessment of what was technically feasible within the constraints of available data, computational resources, and time. Continuous sign segmentation remains an unsolved research problem for single-camera landmark-based systems [9] [5], and attempting it would have produced an unreliable system that obscured the more tractable and

more interesting question of what makes isolated sign recognition generalise to real-world conditions. BSL transfer learning requires a target language dataset of sufficient size to validate the transfer, which does not currently exist in the public domain [1] [3]. The scope discipline that produced the final system, a focused, reliable, fully evaluated ASL isolated sign recogniser with a complete speech pipeline [6] [19], was not a compromise but a deliberate research decision, and the iterative versioning methodology that drove it produced insights that a single ambitious attempt would never have revealed. The most valuable of those insights, that removing features improved performance more than adding architectural complexity [15], would not have been discovered without the systematic comparison across six versions that scope discipline made possible.

6.3 Future Work

Several concrete directions emerge naturally from the findings and limitations of this project. The highest priority is broader community evaluation with native deaf signers, building on the expert evaluation conducted with Professor Adam Schembri and extending it to a wider range of participants from the UK deaf community [3]. This will provide the most meaningful evidence of whether VoiceForAll achieves its core purpose of reducing the communication barrier in real everyday interactions, a question that no amount of technical benchmarking can answer. The second priority is implementing a signer-independent train/test split [1], which would provide a more complete picture of how well the system generalises to new users and enable a more meaningful comparison with the PopSign paper benchmark. The third direction addresses the hands-only limitation directly [5]: selectively reintroducing a small number of facial reference landmarks, specifically the positions of the nose, mouth, chin, and eyes [7], would enable the system to distinguish the subset of signs whose identity depends on facial location, without reintroducing the full pose noise that caused the test-to-real-world gap in earlier versions [5]. The fourth direction is BSL extension through transfer learning [6] from the ASL-trained model, the original ambition of the project and arguably its most impactful potential contribution, as a working ASL-to-BSL transfer learning methodology would provide a path to building recognition systems for sign languages that currently lack sufficient training data [3] [1]. The fifth direction is text-to-sign video playback using the pre-recorded signing videos already available in the PopSign dataset [1], which would complete the bidirectional communication pipeline and allow hearing people to look up how to sign a word by typing it, requiring no generative AI and achievable within a relatively short development timeline [19]. Finally, cross-platform deployment through replacing the macOS-specific text-to-speech component with a platform-independent library would make the system accessible to a significantly wider audience, moving VoiceForAll closer to the genuinely accessible, universally available communication tool that motivated the project from the beginning [3].

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